

Exotics

(1. American Option and Stopping Times)

European Pricing (Recap)

Consider a binomial tree with n periods. Let $V_n = g(S_n)$ be the payoff of a derivative security. For example, for call option the payoff $g(S_n) = (S_n - K)^+$

By backward induction, the price of the European Option at any node is given as

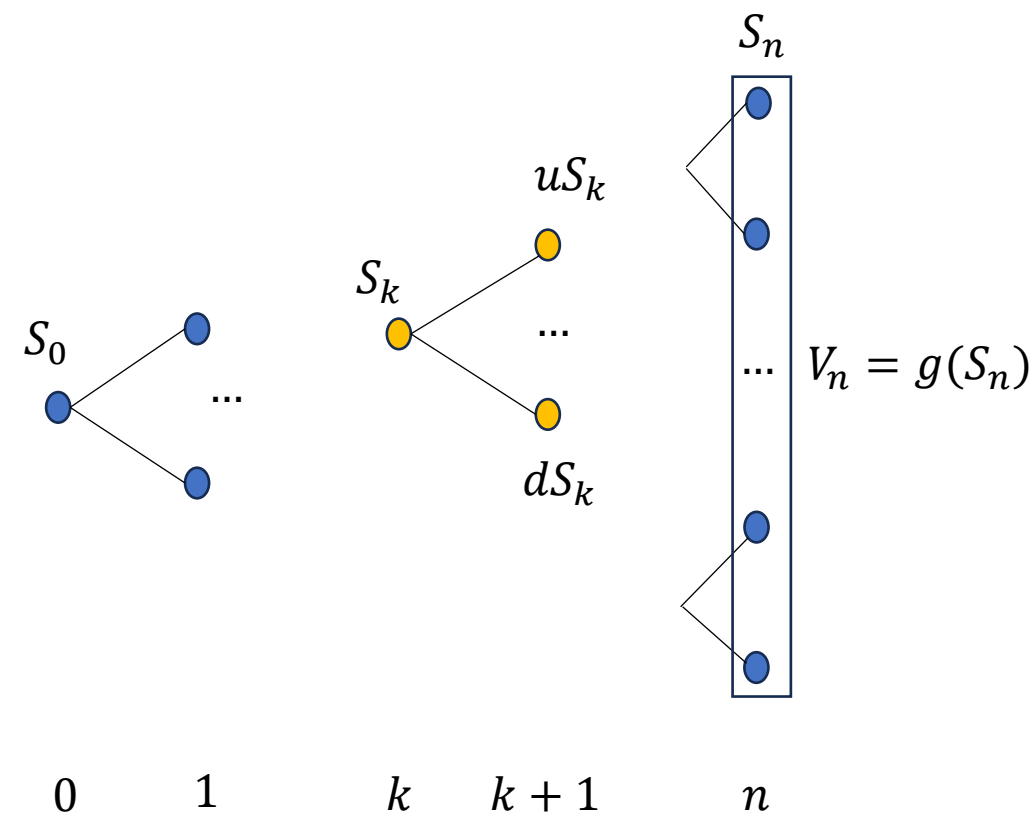
$$v_n(S_n) = g(S_n)$$

$$v_k(S_k) = e^{-r\delta t} [qv_{k+1}(uS_k) + (1 - q)v_{k+1}(dS_k)]$$

The hedging portfolio is given as

$$\Delta_k = \frac{v_{k+1}(uS_k) - v_{k+1}(dS_k)}{S_k(u - d)}$$

(Binomial Tree for Stock)



American Pricing

An American option can be exercised early. This implies the derivative price at any time k is at least the exercise value at that point.

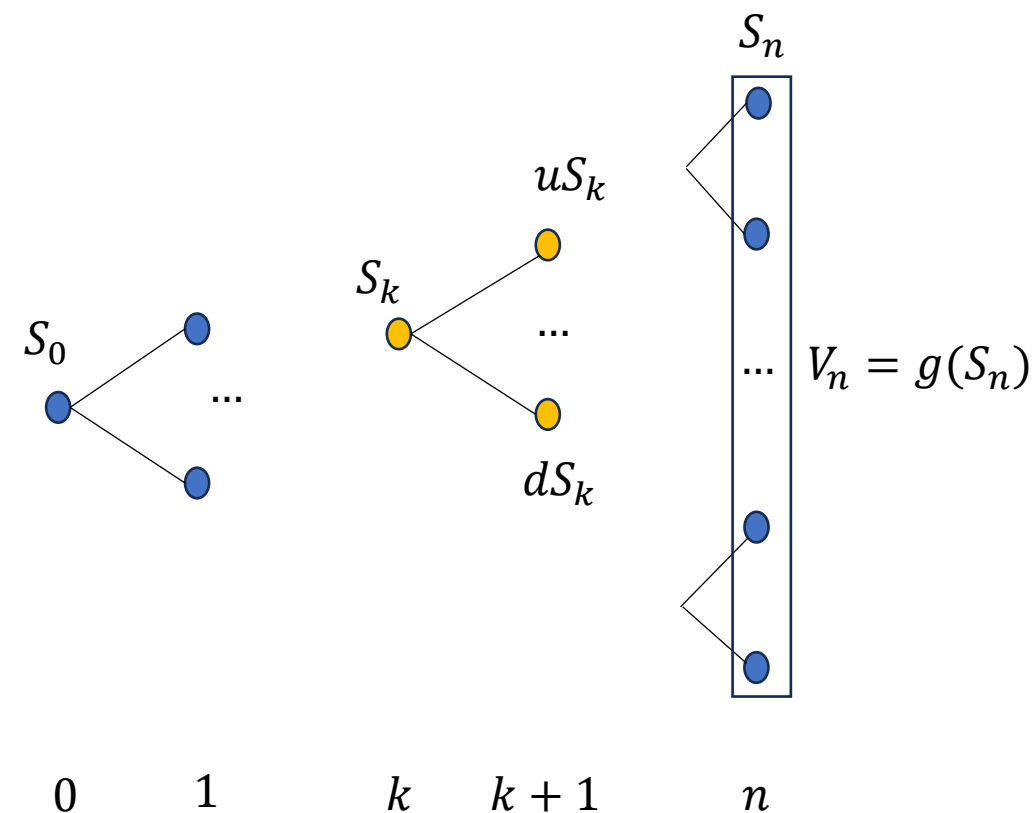
$$v_k(S_k) \geq g(S_k)$$

So, we re-define the backward induction equations as detailed below

$$v_n(S_n) = g(S_n)$$

$$v_k(S_k) = \max \left\{ \underbrace{e^{-r\delta t} [q v_{k+1}(uS_k) + (1-q) v_{k+1}(dS_k)]}_{\text{(continuation value)}}, \underbrace{g(S_k)}_{\text{(Intrinsic value)}} \right\}$$

(Binomial Tree for Stock)



Exercise



Assume the following :

$S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, One period discounting $= \frac{4}{5}$, Risk neutral probabilities : $q = \frac{1}{2}$, Strike : $K = 5$

- ▶ Find the value of a European Put and an American put using a two-period binomial tree ($n = 2$).
- ▶ Find out the hedging portfolio using the formula developed previously.
- ▶ Verify at every node if the replicating portfolio matches the derivative.

Solution



Put Option

$S_0 =$	4
$K =$	5
$DF =$	0.8
$q =$	0.5
$u =$	2
$d =$	0.5

Stock Price Tree

		16
	8	
4		4
	2	
		1

Bond Price Tree

		1.5625
	1.25	
1		1.5625
	1.25	
		1.5625

$v_n(S_n) = g(S_n)$

$v_k(S_k) = e^{-r\delta t}[qv_{k+1}(uS_k) + (1 - q)v_{k+1}(dS_k)]$

European Put

		0
	0.4	
0.96		1
	2	
		4

Hedging Portfolio : Δ

	$\Delta_1(up)$	
Δ_0	-0.0833	
-0.2667		
	-1	
	$\Delta_1(down)$	

Bank account : ψ

	0.85333	
2.02667		
	3.2	

$v_n(S_n) = g(S_n)$

$v_k(S_k) = \max\{e^{-r\delta t}[qv_{k+1}(uS_k) + (1 - q)v_{k+1}(dS_k)], g(S_k)\}$

American Put

		0
	0.4	
1.36		1
	3	
		4

Hedging Portfolio : Δ

	$\Delta_1(up)$	
Δ_0	-0.0833	
-0.4333		
	-1	
	$\Delta_1(down)$	

Bank account : ψ

	0.85333	
3.09333		
	4	

Hedging Portfolio for American Option

Because, American options can be exercised early, the self-financing replicating portfolio is not exact. Instead, the dynamics for replicating portfolio for American options is given as

$$\pi_{k+1} = \Delta_k S_{k+1} + \left(\frac{B_{k+1}}{B_k} \right) (\pi_k - \Delta_k S_k - C_k)$$
$$\pi_{k+1} = \left(\frac{B_{k+1}}{B_k} \right) \pi_k + \Delta_k \left(S_{k+1} - \left(\frac{B_{k+1}}{B_k} \right) S_k \right) - \left(\frac{B_{k+1}}{B_k} \right) C_k$$

Where $C_k \geq 0$ is the amount consumed.

Note that discounted portfolio process is a **super-martingale** under risk-neutral measure.

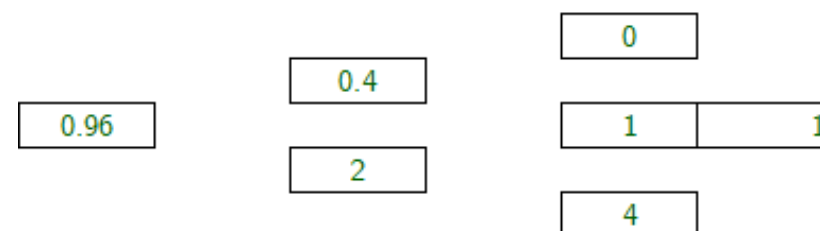
$$\mathbb{E}^{\mathbb{Q}} \left(\frac{\pi_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k \right) = \frac{\pi_k}{B_k} + \Delta_k \left(\mathbb{E}^{\mathbb{Q}} \left(\frac{S_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k \right) - \frac{S_k}{B_k} \right) - \frac{C_k}{B_k} = \frac{\pi_k}{B_k} - \frac{C_k}{B_k} \leq \frac{\pi_k}{B_k}$$

What is C_k ?

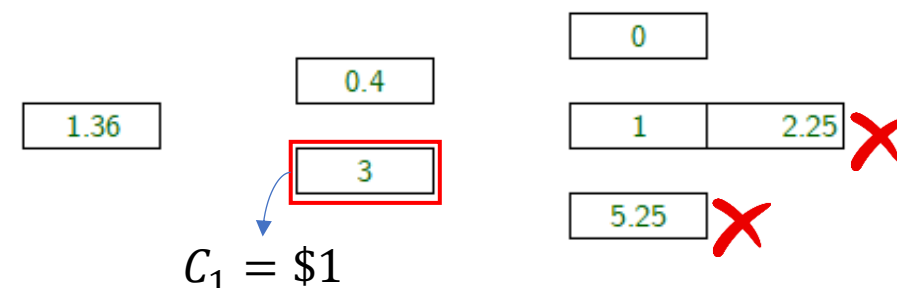
Whenever the American option is early-exercised, the value in that node will be higher than the continuation value. This is where the process **stops**. If we naively continue the risk-free growth from this point, it will result in an overestimated value of the derivative in the future nodes. Because of early exercise, the exact self-financing replication strategy breaks.

Whenever the **intrinsic value** > **continuation value**, if the option is not exercised (= continued), then the seller will consume the wealth C_k . Notice in our example, $C_1 = \$1$ for the node where the American option value exceeds the European counterpart by \$ 1.

(European Put)

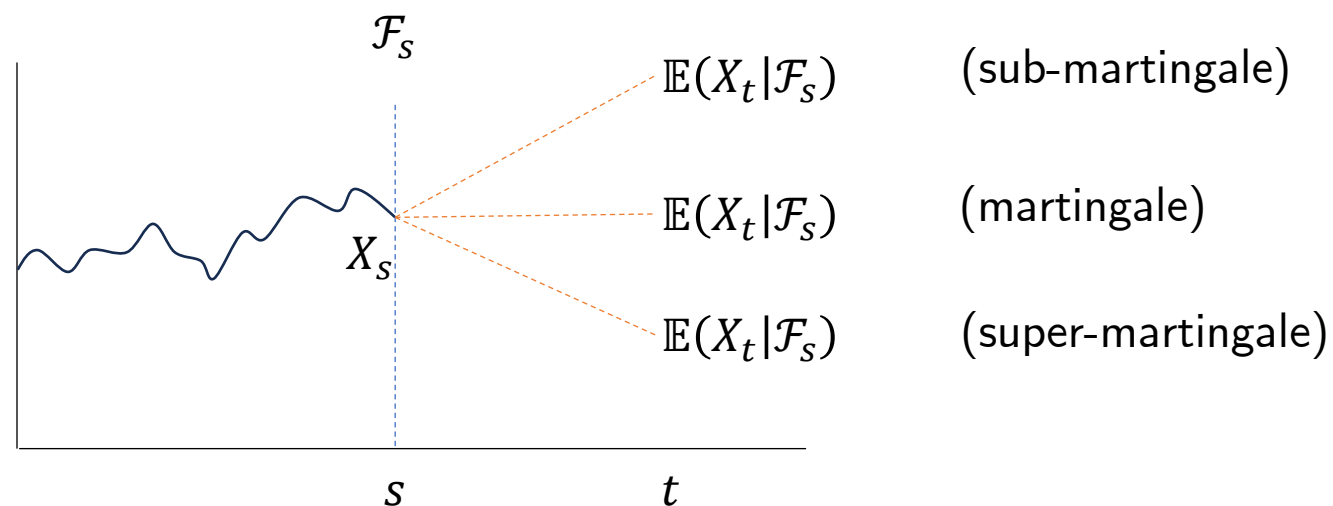


(American Put)



Martingale, Sub-Martingale, Super-Martingale

Below are the definitions of Martingales, Sub-Martingales and Super-Martingales



Martingale : $\mathbb{E}(X_t | \mathcal{F}_s) = X_s$ (no trend, drift=0)

Sub-Martingale : $\mathbb{E}(X_t | \mathcal{F}_s) \geq X_s$ (trending up, drift $\geq$ 0)

Super-Martingale : $\mathbb{E}(X_t | \mathcal{F}_s) \leq X_s$ (trending down, drift $\leq$ 0)

Stopping-Time

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space with filtration $\{F_k\}_{k=0}^n$. A **stopping time** is a random variable which is defined as $\tau: \Omega \rightarrow \{0, 1, 2, \dots, n\} \cup \infty$ with the property that

$$\{\omega \in \Omega; \tau(\omega) = k\} \in \mathcal{F}_k, \forall k = 0, 1, 2, \dots, n, \infty$$

In other words, τ is a stopping time if we know to stop at $\tau = t$ by using the information available till t . We do not need to look into the future. We can also say stopping time is $\mathcal{F}_t$ -adapted. Stopping time refers to a random time when we stop observing a process based on some rule.

Applications of stopping times

- ▶ American Option (Exercise time)
- ▶ Barrier Option (First passage times or hitting times)
- ▶ Credit Risk (firm value drops under a certain threshold)

Examples of Stopping time

Consider a simple symmetric random walk process $\{Z_t\}$ with $Z_0 = 0$

$$Z_t = \sum_{i=1}^n X_i \quad X_i = \begin{cases} 1, & \text{prob} = 0.5 \\ -1, & \text{prob} = 0.5 \end{cases}$$

Which of the following is a stopping time ?

▶ I) $T_1 = \min \{t: Z_t = 10\}$ ✓

▶ II) $T_2 = \min \{t: |Z_t| = 10\}$ ✓

▶ III) $T_3 = \max \{t: |Z_t| = 10\}$ ✗

Important Results

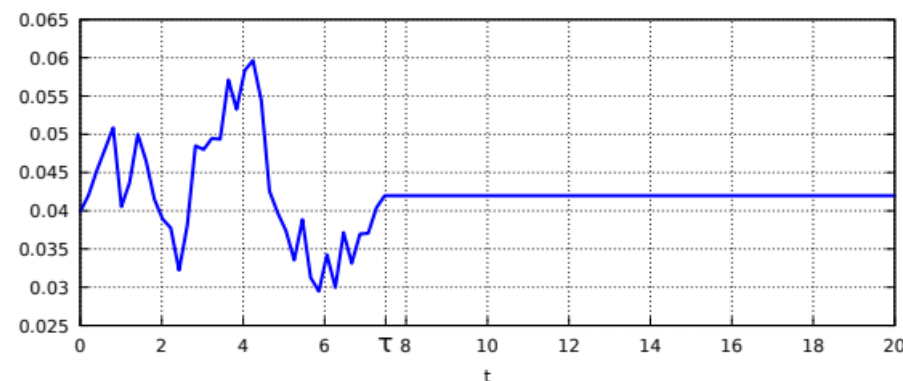
Every constant time is trivially stopping time. In addition, if τ and θ are two stopping times, then

- 1) The minimum $\tau \wedge \theta := \min(\tau, \theta)$ is also a stopping time
- 2) The maximum $\tau \vee \theta := \max(\tau, \theta)$ is also a stopping time

A **stopped process** $Z_{t \wedge \tau}$ is defined as

$$Z_t^\tau = Z_{t \wedge \tau} = \begin{cases} Z_t, & t \leq \tau \\ Z_\tau, & t > \tau \end{cases}$$

Using indicator function we can also write $Z_{t \wedge \tau} = Z_t 1_{\{t \leq \tau\}} + Z_\tau 1_{\{t > \tau\}}$



Martingale with Stopping time

Consider a simple symmetric random walk process $\{Z_t\}$ with $Z_0 = 0$. We know that Z_t is a martingale for any deterministic time t , i.e.,

$$\mathbb{E}[Z_t] = \mathbb{E}[Z_0]$$

But how about random stopping times ?

► I) $\tau = \min \{t: |Z_t| = 10\}$

► II) $\rho = \min \{t: Z_t = 10\}$

Doob's Optional Stopping Theorem (OST)

When random times are involved, we cannot automatically enforce the martingale property. There are certain regularity conditions which need to be met. It's called '**Doob's Optional Stopping**' theorem. The theorem states that

If τ is a random stopping time, then $\mathbb{E}[Z_\tau] = \mathbb{E}[Z_0]$ is true if any of the following conditions hold

- ▶ There exists a constant c such that for all t , $|Z_t| < c$ a.s Or
- ▶ There exists a constant c such $\tau < c$, a.s Or
- ▶ There exists a constant c such for all t , $\mathbb{E}[|Z_{t+1} - Z_t| \mathcal{F}_t] < c$ and $\mathbb{E}[\tau] < \infty$ a.s

To verify these conditions in our previous example, consider the stopped process Z_t^τ .

$$Z_t^\tau = Z_{t \wedge \tau} = \begin{cases} Z_t, & t \leq \tau \\ Z_\tau, & t > \tau \end{cases}$$

Stopping time in American Option (I)

Let's define stopping time $\tau(\omega) = \min\{k ; v_k(S_k) = (5 - S_k)^+\}$ which denotes the first time the option value agrees with the intrinsic value. Also, we will use the short-hand for the events

$$A_u = \{\text{first move is up}\} = \{uu, ud\}$$

$$A_d = \{\text{first move is down}\} = \{du, dd\}$$

We note that
$$\tau(\omega) = \begin{cases} 1, & \text{if } \omega \in A_d \\ 2, & \text{if } \omega \in A_u \end{cases}$$

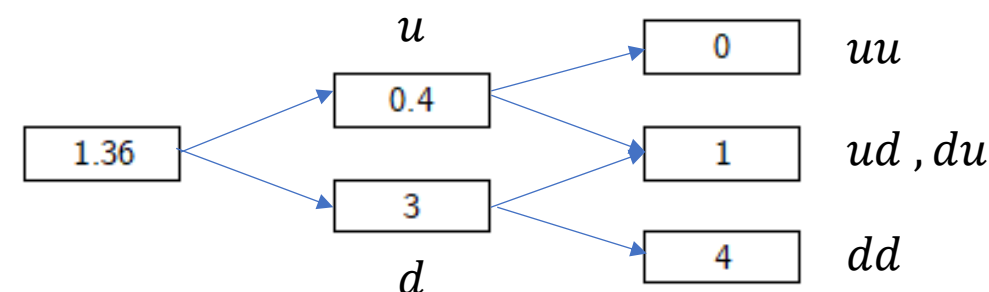
Also, we verify that τ is indeed a stopping time.

$$\{\omega ; \tau(\omega) = 0\} = \phi \in \mathcal{F}_0$$

$$\{\omega ; \tau(\omega) = 1\} = A_d \in \mathcal{F}_1$$

$$\{\omega ; \tau(\omega) = 2\} = A_u \in \mathcal{F}_2$$

(Binomial Tree for Option)



$$\Omega = \{uu, ud, du, dd\}$$

$$\mathcal{F}_0 = \{\phi, \Omega\} \quad \mathcal{F}_1 = \{\phi, \Omega, A_u, A_d\}$$

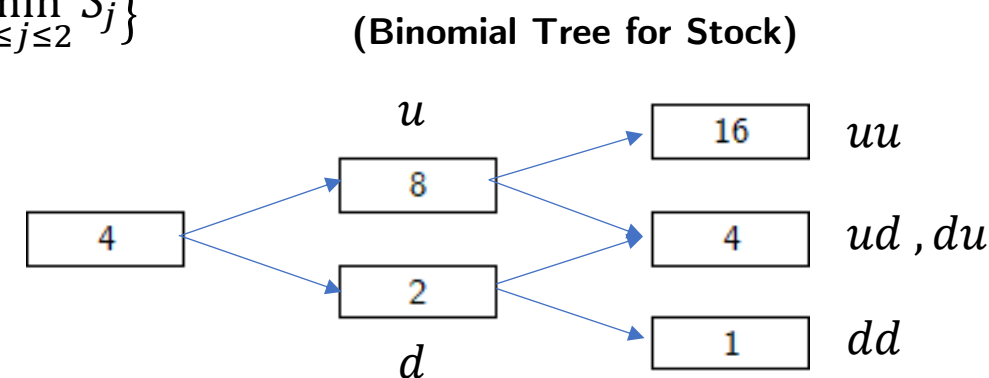
$$\mathcal{F}_2 = \left\{ \begin{array}{l} \phi, \Omega, A_u, A_d, uu, ud, du, dd, \text{ and} \\ \text{all sets that can be created taking unions of these} \end{array} \right\}$$

Stopping time in American Option (II)

Let's look at another random time on the tree that is NOT a stopping time. Define,

$$\rho(\omega) = \min\{k ; S_k = \min_{0 \leq j \leq 2} S_j\}$$

We note that
$$\rho(\omega) = \begin{cases} 0, & \text{if } \omega \in A_u \\ 1, & \text{if } \omega = du \\ 2, & \text{if } \omega = dd \end{cases}$$



Also, we verify that ρ is NOT a stopping time.

$$\{\omega ; \rho(\omega) = 0\} = A_u \notin \mathcal{F}_0$$

$$\{\omega ; \rho(\omega) = 1\} = du \notin \mathcal{F}_1$$

$$\{\omega ; \rho(\omega) = 2\} = dd \in \mathcal{F}_2$$

$$\Omega = \{uu, ud, du, dd\}$$

$$\mathcal{F}_0 = \{\phi, \Omega\} \quad \mathcal{F}_1 = \{\phi, \Omega, A_u, A_d\}$$

$$\mathcal{F}_2 = \left\{ \begin{array}{l} \phi, \Omega, A_u, A_d, uu, ud, du, dd, \text{ and} \\ \text{all sets that can be created taking unions of these} \end{array} \right\}$$

Optional Sampling Theorem



The Optional Sampling theorem uses the same idea but for two random stopping times. So, if τ and ρ are two **bounded** stopping times, i.e., $\tau \leq c$, $\rho \leq c$, a.s where c is a constant, and

If $\tau \leq \rho$ a.s then,

$$Z_\tau = \mathbb{E}(Z_\rho | \mathcal{F}_\tau) \quad (\text{martingale})$$

$$Z_\tau \leq \mathbb{E}(Z_\rho | \mathcal{F}_\tau) \quad (\text{sub-martingale})$$

$$Z_\tau \geq \mathbb{E}(Z_\rho | \mathcal{F}_\tau) \quad (\text{super-martingale})$$

Optional Sampling Theorem on the Tree

To illustrate the idea of Optional Sampling theorem on the tree we define two topping times

$$\tau(\omega) = \min\{k ; v_k(S_k) = (5 - S_k)^+\} \quad \rho(\omega) = 2$$

Under risk neutral probabilities the discounted stock price $Z_k = \left(\frac{4}{5}\right)^k S_k$ is a martingale.

Verify that for all paths ω

$$Z_\tau = \mathbb{E}(Z_\rho | \mathcal{F}_\tau)$$

(in other words)

$$\left(\frac{4}{5}\right)^\tau S_\tau = \mathbb{E}\left(\left(\frac{4}{5}\right)^2 S_2 \middle| \mathcal{F}_\tau\right)$$